

Data Structures Lab (CL2001)

Date: October 16th 2025

Instructor(s)

Mr. Muhammad Kamran

Sessional Exam

Total Time (Hrs): 2 hrs

Total Marks: 20

Total Questions: 2

Roll No

Section

Student Signature

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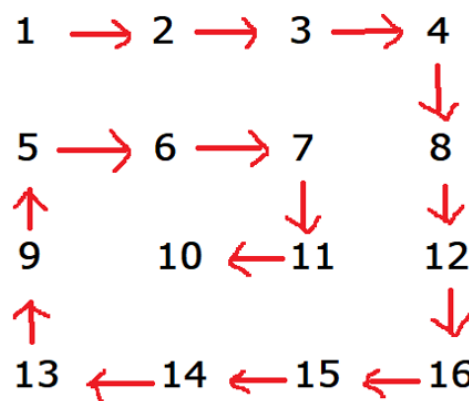
Attempt all the questions.

CLO #: 1

Q1: [11 Marks]

Write a **recursive program** to print all elements of a given 2D matrix in **clockwise spiral order**.
Your solution must not use any loops for traversal

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16



- a) [2 Marks] Recursive function to handle left to right printing
void printRowLeftToRight(int matrix[][MAX], int row, int colStart, int colEnd)
- b) [2 Marks] Recursive function to handle top to bottom printing
void printColumnTopToBottom(int matrix[][MAX], int col, int rowStart, int rowEnd)
- c) [2 Marks] Recursive function to handle right to left printing
void printRowRightToLeft(int matrix[][MAX], int row, int colStart, int colEnd)
- d) [2 Marks] Recursive function to handle bottom to top printing
void printColumnBottomToTop(int matrix[][MAX], int col, int rowStart, int rowEnd)
- e) [3 Marks] Use part a to d to implement spiral traversal
void printSpiralRecursively(int matrix[][MAX], int top, int bottom, int left, int right)

CLO #: 2

Q2: [9 Marks]

- a) [4 Marks] You are required to write a program to **find the Kth node from the end** of a **singly linked list** using **only one traversal** of the list.
- b) [4 Marks] Write an **iterative function** moveLastToFront(Node* head) that moves the **last node** of a singly linked list to the **front**, keeping the order of all other nodes **unchanged**. You are **not allowed to**

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use extra data structures such as arrays, stacks, or queues. The operation must be done **in-place** using pointer manipulation only.

- c) [1 Mark] Create a Menu for all questions